



MAHEDI HASSAN

BIOCHEMISTRY & MOLECULAR BIOLOGY | COMPUTATIONAL DRUG DESIGN

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PROFESSIONAL SUMMARY

Dedicated and research-oriented Biochemistry and Molecular Biology professional with extensive experience in computational drug design, molecular dynamics simulation, and laboratory operations. Proven track record in academia as a Lecturer in Biochemistry, coupled with research assistantship in computational drug design. Seeking opportunities to contribute to scientific research and innovation in bioinformatics and molecular biology.

PROFESSIONAL EXPERIENCE

Lecturer in Biochemistry

Jan 2023 – Present

Ideal MATS & PIMT, Bangladesh

Delivering engaging lectures on Biochemistry and Molecular Biology, evaluating student performance, and facilitating academic growth.

Lecturer in Biochemistry & Molecular Biology

Oct 2021 – Dec 2023

Bhawal Gazipur Medical Institute (BGMI)

Conducted lectures, designed assessments, and provided constructive feedback to students.

Biochemist (Part-Time)

Duration: 1.9 Years

A.K. AL-Sheefa Hospital & Diagnostic Center

Managed laboratory operations, conducted diagnostic tests, and contributed to clinical research initiatives.

EDUCATIONAL QUALIFICATION

Bachelor of Science (B.Sc.) in Biochemistry & Molecular Biology

2017 – 2021

Gono Bishwabidyalay | Savar, Dhaka

CGPA: 3.71 / 4.00

Diploma in Laboratory Medicine

2012 – 2016

Chandana Institute of Medical Technology (CIMT)

GPA: 3.81 / 4.00 (Higher Secondary Level) | Focused learning and practical training in laboratory medicine.

Secondary School Certificate (SSC) - Science

2006 – 2012

M. E. H Arif College | Konabari, Gazipur

GPA: 4.44 / 5.00

RESEARCH FOCUS & PROJECTS

In Silico Based Drug Design and Discovery: Utilizing computational approaches including molecular docking, virtual screening, and structure-based drug design against various disease targets.

AMDET & Toxicity Prediction: Exploring bioactive phytochemicals from medicinal plants employing computational ADMET prediction and molecular dynamics simulations.

Nanoparticle Based Drug Design: Investigating nanoparticle-based drug delivery systems for targeted therapy.

Docking Automation Project: Developed automated pipelines for Molecular Docking, Virtual Screening, and ADMET analysis using Python.

Carbon Deposition Project: Detailed visualization and analysis of carbon deposition processes exploring structural interactions using advanced simulation techniques.

CRISPR Cas9 Project (Guided RNA): Analysis of Guided RNA (gRNA) efficiency and target specificity for the CRISPR-Cas9 system.

PUBLICATIONS

Effects of Supplementing Layers with Varying Levels of Alfalfa and Moringa Fresh Leaves on Egg Production and Quality

Alemayehu Guteta, Mahedi Hassan, Khokon Hossain, Mammo Mengesha Erdaw (Published Dec 2023)

Identification, Pathogenicity, Public Health and Economic Importance of Tapeworms of Poultry

Daniel Tamirat, Mahedi Hassan, Saima Akter Bristy, Md Khokon Hossain, Md. Raquibul Islam (Published Feb 2024)

SKILLS & EXPERTISE

- Computational Skills:** MD Simulation, Molecular Docking, Virtual Screening, ADMET Prediction, AI-Assisted Drug Design, Bioinformatics Tools, Data Analysis & Visualization.
- Software Expertise:** GROMACS, AMBER, Origin Pro, AutoDock Vina, Discovery Studio, PyMOL, BioPython, HTML.
- Laboratory Techniques:** Cell Culture, Microscopy & Imaging, Protein Expression, ELISA & Spectroscopy, Chromatography & MS, DNA/RNA Extraction.
- Soft Skills:** Academic Writing, Research Leadership, Team Collaboration, Problem Solving, Teaching & Mentoring.
- Languages:** Bangla (Native), English : Fluent (Medium of Instruction in B.Sc. was English).

TRAINING & CERTIFICATIONS

- Biosafety & Biosecurity (Certified Trainer)** – Bangladesh Biosafety and Biosecurity Society
- Student Research Ambassador** – National Student Research Institute (NSRI)
- Bioinformatics & Drug Design** – Dawn of Bioinformatics Limited
- Advanced English Speaker (B2 Level)** – British Council, Bangladesh
- Computer Office Application Specialist** – BTEB
- Medical Campaign Organizer** – Bhawal Gazipur Medical Institute
- Medication Safety for NTDs** – World Health Organization (WHO)

RESEARCH COLLABORATION

- Research Collaboration**

Chulalongkorn University

Collaborating on a project focused on nanoparticle-based cyclodextrin encapsulation to develop a slow and sustained drug delivery system.
- Research Collaboration**

Genexa Foundation, Riyadh, Saudi Arabia

Supported computational drug design projects and in-silico research methodologies.

REFERENCES

- Dr. Fuad Hossain**
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